

Project Synopsis Template

1. Title

AI-Powered Crop Recommendation System Using Machine Learning

Team Details

Provide details of all four team members. The first student will be the Team Lead.

Student Name	College Name	Branch	Mobile No	Mail ID	Role (Lead/Member)
Tammireddy Deepika	International School of Technology and Sciences for Women	Computer Science Engineering	9948814643	deepuists23@gmail.com	Team Leader
Gelam PujithaRani	International School of Technology and Sciences for Women	Computer Science Engineering	8790519834	pujitharanigelam@gmail.com	Team Member1
Soyam Chandana	International School of Technology and Sciences for Women	Computer Science Engineering	6301227029	soyamchandhana@gmail.com	Team Member2
Koncha Likitha	International School of Technology and Sciences for Women	Computer Science Engineering	9912273149	likithakoncha05@gmail.com	Team Member3

Program Track and Domain

Program Track (Engg-HUB / Engg-Spoke / Science-HUB / Science-Spoke / Commerce-Spoke)	Domain (Analytics / Machine Learning / Deep Learning / GenAI)
Engg-Spoke	Machine Learning

Guide Details

Bharathi Raja M

2. Problem Statement

Agriculture plays a vital role in food production and economic growth. Farmers often face challenges in selecting the most suitable crop due to variations in soil nutrients and environmental conditions. Traditional crop selection methods may result in lower productivity and financial losses. This project aims to develop an AI-powered Crop Recommendation System using Machine Learning that analyzes soil nutrients (Nitrogen, Phosphorus, Potassium) and environmental factors such as temperature, humidity, pH, and rainfall to recommend the most suitable crop. This system helps farmers make informed decisions and improve agricultural productivity.

3. Objectives

1. To analyze agricultural and soil data.
2. To build a machine learning model for crop recommendation.
3. To recommend suitable crops based on environmental conditions.
4. To improve decision-making in agriculture.
5. To achieve accurate crop prediction using machine learning.

4. Methodology / Implementation Plan

Workflow

1. Data Collection
2. Data Preprocessing
3. Feature Selection
4. Model Training

5. Model Evaluation

6. Crop Prediction

Algorithm Used

- Random Forest Classifier

Tools & Technologies

- Python
- Pandas
- Scikit-learn
- Matplotlib
- Google Colab
- GitHub

5. Resources and Timeline

A. Hardware & Software Requirements

Hardware

- Laptop / PC
- Minimum 4GB RAM

Software

- Windows 10/11
- Python
- Google Colab
- GitHub

B. Dataset

Dataset Availability

Available:

https://drive.google.com/file/d/1pFiioC7t_vLGfxTBUoOXZFy9qPWX8pip/view?usp=drive_link

Dataset Source

Dataset: Crop Recommendation Dataset

Dataset Size: 2200 Records

Features:

- Nitrogen (N)
- Phosphorus (P)
- Potassium (K)
- Temperature
- Humidity
- pH
- Rainfall
- Crop Label

C. Project Timeline

Phase	Task	Completion Date
Phase 1	Dataset Collection	Completed
Phase 2	Data Preprocessing	Completed
Phase 3	Model Development	Completed
Phase 4	Testing & Evaluation	Completed
Phase 5	Documentation & PPT	Completed

6.Expected Outcomes / Conclusion

The project successfully develops a Machine Learning-based Crop Recommendation System capable of recommending suitable crops based on soil and environmental parameters. The Random Forest model achieved an accuracy of 99.32%, demonstrating high prediction performance. The system can assist farmers in selecting appropriate crops and improving agricultural productivity.

7. References

1. Crop Recommendation Dataset (Kaggle)
2. Python Documentation
3. Scikit-learn Documentation
4. Google Colab Documentation
5. Random Forest Classifier Documentation
6. GitHub Repository of the Project
7. <https://github.com/TammireddyDeepika/AI-Powered-Crop-Recommendation-System-Using-Machine-Learning>